V3 empirical study · Instruction

Six models · three underlyings · twelve expiry windows · 1-day hourly calibration

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of six American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and the twelve expiry windows?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Mark column ranks 1–6 (1 = lowest RMSE%).

Experimental design (held fixed for every model)

- Models: GBM, GARCH(1,1), Heston (no jumps), Merton jump-diffusion, GARCH–Merton, Heston–Merton (Bates).
- Companies: SPY (primary), AAPL, MSFT. Each name is calibrated and priced separately.
- Windows: twelve monthly third-Friday expiries (Oct 2022–Sep 2023). Evaluation is the Friday immediately before expiry, not the expiry Friday itself.
- 1-day study = that Friday. 7-day study = five NYSE sessions ending that Friday. This report is the 1-day study: that Friday only (seven hourly stamps 09:59–15:59).
- Calibration window: 1-day of 1-minute RTH equity returns and listed quotes ending at each hourly update (09:59–15:59). Parameters estimated at t use only data with timestamp ≤ t (no look-ahead).
- If history is shorter than the requested lookback (first hours of 10 Oct 2022), the estimator uses all available 1-minute observations up to t (equity file starts 2022-09-30).

V3 empirical study · Instruction (continued)

Six models · three underlyings · twelve expiry windows · 1-day hourly calibration

Shared market sample and filters (not model-specific)

- For each company × window, one listed-call sample is drawn once and reused by all six models: nearest-ATM call as-of each hourly evaluation timestamp (09:59, 10:59, 11:59, 12:59, 13:59, 14:59, 15:59), DTE 7–60, listed expiry.
- LSM does not run on a 15-minute or 5-minute quote grid. Friday 15:59 is listed as a scheduled stamp but remaining_horizon = 0, so LSM is skipped (at least two MC steps required). RMSE% uses only timestamps with a valid remaining horizon.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite S, K, C; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq \mathrm{DTE} \leq 60$; (4) $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05, valid bid–ask with relative spread ≤ 50% when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston–Merton: (κ, θ, ξ, ρ, v_0) are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces $\mu \to r$.
- Euler ordering: the stock increment over $[t, t+\Delta t]$ uses the current variance v_t; $v_{t+\Delta t}$ is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud (n_paths = 10000, seed 42). n_steps = remaining MC steps from t to window Friday 15:59, not listed DTE. $\Delta t = 1$ minute(s). Paths are not averaged before stopping. Do not extend paths to listed expiry.

Metrics

- Primary: $\mathrm{RMSE\%} = 100 \times \sqrt{\mathrm{mean}(((C_{model} - C_{mkt}) / C_{mkt})^2)}$. Ranking uses this number only.
- Secondary: $\mathrm{MAE} = \mathrm{mean}\,|C_{model} - C_{mkt}|$ (dollars). Bias \$ = $\mathrm{mean}(C_{model} - C_{mkt})$ (dollars). $\mathrm{Bias\%} = 100 \times \mathrm{mean}((C_{model} - C_{mkt})/C_{mkt})$. Positive bias = model expensive vs market. Early-exercise fraction = mean share of LSM paths that stop before the window end.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all six models)

Window	Eval Friday	Sessions	SPY n	AAPL n	MSFT n
Oct 2022 2022-10-21	2022-10-14	1	6	6	6
Nov 2022 2022-11-18	2022-11-11	1	6	6	6
Dec 2022 2022-12-16	2022-12-09	1	6	6	6
Jan 2023 2023-01-20	2023-01-13	1	6	6	6
Feb 2023 2023-02-17	2023-02-10	1	6	6	6
Mar 2023 2023-03-17	2023-03-10	1	6	6	6
Apr 2023 2023-04-21	2023-04-14	1	6	6	6
May 2023 2023-05-19	2023-05-12	1	6	6	6
Jun 2023 2023-06-16	2023-06-09	1	6	6	6
Jul 2023 2023-07-21	2023-07-14	1	6	6	6
Aug 2023 2023-08-18	2023-08-11	1	6	6	6
Sep 2023 2023-09-15	2023-09-08	1	6	6	6

Filter funnel · last-step n in the 1-day lookback ending at each window (SPY processed calls)

Window	Input*	No-arb	DTE 7–60	|S/K−1|≤10%	Liquidity	Eval n
Oct22	730	730	730	724	724	6
Nov22	872	872	872	841	841	6
Dec22	547	547	547	539	539	6
Jan23	548	548	548	542	542	6
Feb23	547	547	547	538	538	6
Mar23	591	591	591	585	585	6
Apr23	521	521	521	512	512	6
May23	470	470	470	466	466	6
Jun23	475	475	475	462	462	6
Jul23	548	548	548	535	535	6
Aug23	564	564	564	552	552	6
Sep23	508	508	508	500	500	6

*Input is the processed call panel restricted to the 1-day 1-minute lookback ending at window Friday 15:59. Same filters for AAPL/MSFT. No look-ahead: hourly parameters at t use only data ≤ t. LSM horizon

Page 4 · Table block · SPY · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1-day lookback, hourly rolling, Δt = 1 min, n_paths=10000, seed=42. LSM horizon = remaining time to

Table SPY-1 SPY · Oct 2022 listed expiry 2022-10-21
2022-10-14 09:30 → 2022-10-14 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	41.86	5.9376	-5.9376	-37.80	92.9%	2
GARCH	42.00	5.9640	-5.9640	-37.96	96.2%	4
Heston	42.05	6.0098	-6.0098	-38.25	97.2%	5
Merton	41.96	5.9472	-5.9472	-37.86	95.1%	3
GARCH–Merton	**41.39**	**5.8648**	**-5.8648**	**-37.33**	**90.4%**	**1**
Heston–Merton	42.09	5.9976	-5.9976	-38.18	94.0%	6

Table SPY-2 SPY · Nov 2022 listed expiry 2022-11-18
2022-11-11 09:30 → 2022-11-11 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	**100.00**	**10.0600**	**-10.0600**	**-100.00**	**0.0%**	**1**
GARCH	100.00	10.0600	-10.0600	-100.00	0.0%	2
Heston	100.00	10.0600	-10.0600	-100.00	0.0%	3
Merton	100.00	10.0600	-10.0600	-100.00	0.0%	4
GARCH–Merton	100.00	10.0600	-10.0600	-100.00	0.0%	5
Heston–Merton	100.00	10.0600	-10.0600	-100.00	0.0%	6

Table SPY-3 SPY · Dec 2022 listed expiry 2022-12-16
2022-12-09 09:30 → 2022-12-09 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	100.00	9.0400	-9.0400	-100.00	0.0%	5
GARCH	100.00	9.0400	-9.0400	-100.00	0.0%	4
Heston	99.99	9.0390	-9.0390	-99.99	0.0%	2
Merton	100.00	9.0400	-9.0400	-100.00	0.0%	6
GARCH–Merton	100.00	9.0399	-9.0399	-100.00	0.0%	3
Heston–Merton	**99.99**	**9.0390**	**-9.0390**	**-99.99**	**0.1%**	**1**

Table SPY-4 SPY · Jan 2023 listed expiry 2023-01-20
2023-01-13 09:30 → 2023-01-13 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	**100.00**	**8.1200**	**-8.1200**	**-100.00**	**0.0%**	**1**
GARCH	100.00	8.1200	-8.1200	-100.00	0.0%	2
Heston	100.00	8.1200	-8.1200	-100.00	0.0%	3
Merton	100.00	8.1200	-8.1200	-100.00	0.0%	4
GARCH–Merton	100.00	8.1200	-8.1200	-100.00	0.0%	5
Heston–Merton	100.00	8.1200	-8.1200	-100.00	0.0%	6

RMSE% is 100×√mean(((model−market)/market)²). MAE and Bias $ are in option-premium dollars. Bias% is 100×mean((model−market)/market). Early-ex. frac. is the mean LSM share of paths exercised b

Page 5 · Table block · SPY · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1-day lookback, hourly rolling, Δt = 1 min, n_paths=10000, seed=42. LSM horizon = remaining time to

Table SPY-5 SPY · Feb 2023 listed expiry 2023-02-17
2023-02-10 09:30 → 2023-02-10 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	100.00	9.1600	-9.1600	-100.00	0.0%	2
GARCH	100.00	9.1600	-9.1600	-100.00	0.0%	3
Heston	100.00	9.1600	-9.1600	-100.00	0.0%	4
Merton	100.00	9.1600	-9.1600	-100.00	0.0%	5
GARCH–Merton	**100.00**	**9.1600**	**-9.1600**	**-100.00**	**0.0%**	**1**
Heston–Merton	100.00	9.1600	-9.1600	-100.00	0.0%	6

Table SPY-6 SPY · Mar 2023 listed expiry 2023-03-17
2023-03-10 09:30 → 2023-03-10 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	91.59	8.3781	-8.3781	-91.26	16.7%	4
GARCH	91.83	8.3959	-8.3959	-91.46	16.3%	6
Heston	88.90	8.1107	-8.1107	-88.35	23.2%	2
Merton	91.71	8.3899	-8.3899	-91.39	19.6%	5
GARCH–Merton	91.13	8.3291	-8.3291	-90.73	18.0%	3
Heston–Merton	**88.50**	**8.0720**	**-8.0720**	**-87.93**	**22.6%**	**1**

Table SPY-7 SPY · Apr 2023 listed expiry 2023-04-21
2023-04-14 09:30 → 2023-04-14 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	100.00	7.7900	-7.7900	-100.00	0.0%	3
GARCH	100.00	7.7900	-7.7900	-100.00	0.0%	4
Heston	100.00	7.7897	-7.7897	-100.00	0.0%	2
Merton	100.00	7.7900	-7.7900	-100.00	0.0%	5
GARCH–Merton	100.00	7.7900	-7.7900	-100.00	0.0%	6
Heston–Merton	**99.99**	**7.7896**	**-7.7896**	**-99.99**	**0.0%**	**1**

Table SPY-8 SPY · May 2023 listed expiry 2023-05-19
2023-05-12 09:30 → 2023-05-12 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	**100.00**	**6.7800**	**-6.7800**	**-100.00**	**0.0%**	**1**
GARCH	100.00	6.7800	-6.7800	-100.00	0.0%	2
Heston	100.00	6.7800	-6.7800	-100.00	0.0%	3
Merton	100.00	6.7800	-6.7800	-100.00	0.0%	4
GARCH–Merton	100.00	6.7800	-6.7800	-100.00	0.0%	5
Heston–Merton	100.00	6.7800	-6.7800	-100.00	0.0%	6

RMSE% is 100×√mean(((model−market)/market)²). MAE and Bias $ are in option-premium dollars. Bias% is 100×mean((model−market)/market). Early-ex. frac. is the mean LSM share of paths exercised b

Page 6 · Table block · SPY · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1-day lookback, hourly rolling, Δt = 1 min, n_paths=10000, seed=42. LSM horizon = remaining time to

Table SPY-9 SPY · Jun 2023 listed expiry 2023-06-16
2023-06-09 09:30 → 2023-06-09 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	**100.00**	**5.2200**	**-5.2200**	**-100.00**	**0.0%**	**1**
GARCH	100.00	5.2200	-5.2200	-100.00	0.0%	2
Heston	100.00	5.2200	-5.2200	-100.00	0.0%	3
Merton	100.00	5.2200	-5.2200	-100.00	0.0%	4
GARCH–Merton	100.00	5.2200	-5.2200	-100.00	0.0%	5
Heston–Merton	100.00	5.2200	-5.2200	-100.00	0.0%	6

Table SPY-10 SPY · Jul 2023 listed expiry 2023-07-21
2023-07-14 09:30 → 2023-07-14 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	100.00	6.4700	-6.4700	-100.00	0.0%	2
GARCH	100.00	6.4700	-6.4700	-100.00	0.0%	3
Heston	100.00	6.4700	-6.4700	-100.00	0.0%	4
Merton	100.00	6.4700	-6.4700	-100.00	0.0%	5
GARCH–Merton	**100.00**	**6.4699**	**-6.4699**	**-100.00**	**0.0%**	**1**
Heston–Merton	100.00	6.4700	-6.4700	-100.00	0.0%	6

Table SPY-11 SPY · Aug 2023 listed expiry 2023-08-18
2023-08-11 09:30 → 2023-08-11 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	**100.00**	**7.2300**	**-7.2300**	**-100.00**	**0.0%**	**1**
GARCH	100.00	7.2300	-7.2300	-100.00	0.0%	2
Heston	100.00	7.2300	-7.2300	-100.00	0.0%	3
Merton	100.00	7.2300	-7.2300	-100.00	0.0%	4
GARCH–Merton	100.00	7.2300	-7.2300	-100.00	0.0%	5
Heston–Merton	100.00	7.2300	-7.2300	-100.00	0.0%	6

Table SPY-12 SPY · Sep 2023 listed expiry 2023-09-15
2023-09-08 09:30 → 2023-09-08 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	**100.00**	**5.8200**	**-5.8200**	**-100.00**	**0.0%**	**1**
GARCH	100.00	5.8200	-5.8200	-100.00	0.0%	2
Heston	100.00	5.8200	-5.8200	-100.00	0.0%	3
Merton	100.00	5.8200	-5.8200	-100.00	0.0%	4
GARCH–Merton	100.00	5.8200	-5.8200	-100.00	0.0%	5
Heston–Merton	100.00	5.8200	-5.8200	-100.00	0.0%	6

RMSE% is 100×√mean(((model−market)/market)²). MAE and Bias $ are in option-premium dollars. Bias% is 100×mean((model−market)/market). Early-ex. frac. is the mean LSM share of paths exercised b

Page 7 · Table block · AAPL · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1-day lookback, hourly rolling, Δt = 1 min, n_paths=10000, seed=42. LSM horizon = remaining time to

Table AAPL-1 AAPL · Oct 2022 listed expiry 2022-10-21
2022-10-14 09:30 → 2022-10-14 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	72.72	4.8819	-4.8819	-70.75	56.7%	5
GARCH	73.80	4.9551	-4.9551	-71.81	59.7%	6
Heston	**70.82**	**4.7682**	**-4.7682**	**-69.10**	**53.7%**	**1**
Merton	72.69	4.8751	-4.8751	-70.65	55.0%	4
GARCH–Merton	72.34	4.8362	-4.8362	-70.09	57.2%	3
Heston–Merton	71.44	4.7968	-4.7968	-69.52	55.6%	2

Table AAPL-2 AAPL · Nov 2022 listed expiry 2022-11-18
2022-11-11 09:30 → 2022-11-11 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	99.99	5.3496	-5.3496	-99.99	0.1%	5
GARCH	**99.86**	**5.3428**	**-5.3428**	**-99.86**	**0.4%**	**1**
Heston	99.96	5.3479	-5.3479	-99.96	0.3%	4
Merton	99.99	5.3496	-5.3496	-99.99	0.1%	6
GARCH–Merton	99.87	5.3428	-5.3428	-99.87	0.5%	2
Heston–Merton	99.95	5.3476	-5.3476	-99.95	0.3%	3

Table AAPL-3 AAPL · Dec 2022 listed expiry 2022-12-16
2022-12-09 09:30 → 2022-12-09 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	82.45	4.3228	-4.3228	-81.95	41.9%	4
GARCH	82.76	4.3341	-4.3341	-82.16	41.2%	6
Heston	75.58	3.9557	-3.9557	-74.99	41.3%	2
Merton	82.53	4.3271	-4.3271	-82.03	43.1%	5
GARCH–Merton	81.66	4.2730	-4.2730	-81.00	39.6%	3
Heston–Merton	**75.55**	**3.9528**	**-3.9528**	**-74.93**	**37.4%**	**1**

Table AAPL-4 AAPL · Jan 2023 listed expiry 2023-01-20
2023-01-13 09:30 → 2023-01-13 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	99.83	5.6655	-5.6655	-99.83	0.9%	5
GARCH	99.67	5.6560	-5.6560	-99.67	1.4%	2
Heston	99.71	5.6584	-5.6584	-99.71	1.6%	3
Merton	99.84	5.6657	-5.6657	-99.84	1.1%	6
GARCH–Merton	**99.60**	**5.6521**	**-5.6521**	**-99.60**	**1.5%**	**1**
Heston–Merton	99.75	5.6609	-5.6609	-99.75	1.4%	4

RMSE% is 100×√mean(((model−market)/market)²). MAE and Bias $ are in option-premium dollars. Bias% is 100×mean((model−market)/market). Early-ex. frac. is the mean LSM share of paths exercised b

Page 8 · Table block · AAPL · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1-day lookback, hourly rolling, Δt = 1 min, n_paths=10000, seed=42. LSM horizon = remaining time to

Table AAPL-5 AAPL · Feb 2023 listed expiry 2023-02-17
2023-02-10 09:30 → 2023-02-10 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	98.38	5.6321	-5.6321	-98.38	7.0%	4
GARCH	98.73	5.6519	-5.6519	-98.72	4.9%	6
Heston	94.91	5.4318	-5.4318	-94.88	12.8%	2
Merton	98.30	5.6273	-5.6273	-98.29	6.3%	3
GARCH–Merton	98.55	5.6418	-5.6418	-98.55	5.5%	5
Heston–Merton	**94.76**	**5.4227**	**-5.4227**	**-94.72**	**11.8%**	**1**

Table AAPL-6 AAPL · Mar 2023 listed expiry 2023-03-17
2023-03-10 09:30 → 2023-03-10 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	84.86	3.8935	-3.8935	-84.18	31.9%	5
GARCH	83.73	3.8332	-3.8332	-82.88	29.7%	4
Heston	**78.28**	**3.5671**	**-3.5671**	**-77.13**	**33.8%**	**1**
Merton	84.94	3.8979	-3.8979	-84.28	33.2%	6
GARCH–Merton	83.38	3.8154	-3.8154	-82.50	31.2%	3
Heston–Merton	78.31	3.5685	-3.5685	-77.16	32.9%	2

Table AAPL-7 AAPL · Apr 2023 listed expiry 2023-04-21
2023-04-14 09:30 → 2023-04-14 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	96.51	5.3038	-5.3038	-96.43	13.7%	3
GARCH	97.51	5.3608	-5.3608	-97.47	11.7%	6
Heston	96.58	5.3071	-5.3071	-96.49	14.8%	4
Merton	**96.50**	**5.3037**	**-5.3037**	**-96.43**	**13.9%**	**1**
GARCH–Merton	97.14	5.3398	-5.3398	-97.09	12.4%	5
Heston–Merton	96.50	5.3031	-5.3031	-96.42	13.2%	2

Table AAPL-8 AAPL · May 2023 listed expiry 2023-05-19
2023-05-12 09:30 → 2023-05-12 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	100.00	4.2500	-4.2500	-100.00	0.0%	3
GARCH	100.00	4.2500	-4.2500	-100.00	0.0%	4
Heston	100.00	4.2500	-4.2500	-100.00	0.0%	2
Merton	100.00	4.2500	-4.2500	-100.00	0.0%	5
GARCH–Merton	**100.00**	**4.2500**	**-4.2500**	**-100.00**	**0.0%**	**1**
Heston–Merton	100.00	4.2500	-4.2500	-100.00	0.0%	6

RMSE% is 100×√mean(((model−market)/market)²). MAE and Bias $ are in option-premium dollars. Bias% is 100×mean((model−market)/market). Early-ex. frac. is the mean LSM share of paths exercised b

Page 9 · Table block · AAPL · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1-day lookback, hourly rolling, Δt = 1 min, n_paths=10000, seed=42. LSM horizon = remaining time to

Table AAPL-9 AAPL · Jun 2023 listed expiry 2023-06-16
2023-06-09 09:30 → 2023-06-09 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	100.00	3.4500	-3.4500	-100.00	0.0%	5
GARCH	100.00	3.4500	-3.4500	-100.00	0.0%	3
Heston	99.93	3.4477	-3.4477	-99.93	0.3%	2
Merton	100.00	3.4500	-3.4500	-100.00	0.0%	6
GARCH–Merton	100.00	3.4500	-3.4500	-100.00	0.0%	4
Heston–Merton	**99.93**	**3.4475**	**-3.4475**	**-99.93**	**0.2%**	**1**

Table AAPL-10 AAPL · Jul 2023 listed expiry 2023-07-21
2023-07-14 09:30 → 2023-07-14 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	**100.00**	**5.5500**	**-5.5500**	**-100.00**	**0.0%**	**1**
GARCH	100.00	5.5500	-5.5500	-100.00	0.0%	2
Heston	100.00	5.5500	-5.5500	-100.00	0.0%	3
Merton	100.00	5.5500	-5.5500	-100.00	0.0%	4
GARCH–Merton	100.00	5.5500	-5.5500	-100.00	0.0%	5
Heston–Merton	100.00	5.5500	-5.5500	-100.00	0.0%	6

Table AAPL-11 AAPL · Aug 2023 listed expiry 2023-08-18
2023-08-11 09:30 → 2023-08-11 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	**100.00**	**3.4000**	**-3.4000**	**-100.00**	**0.0%**	**1**
GARCH	100.00	3.4000	-3.4000	-100.00	0.0%	2
Heston	100.00	3.4000	-3.4000	-100.00	0.0%	3
Merton	100.00	3.4000	-3.4000	-100.00	0.0%	4
GARCH–Merton	100.00	3.4000	-3.4000	-100.00	0.0%	5
Heston–Merton	100.00	3.4000	-3.4000	-100.00	0.0%	6

Table AAPL-12 AAPL · Sep 2023 listed expiry 2023-09-15
2023-09-08 09:30 → 2023-09-08 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	100.00	3.8250	-3.8250	-100.00	0.0%	6
GARCH	100.00	3.8249	-3.8249	-100.00	0.0%	4
Heston	**99.96**	**3.8234**	**-3.8234**	**-99.96**	**0.2%**	**1**
Merton	100.00	3.8250	-3.8250	-100.00	0.0%	5
GARCH–Merton	99.99	3.8246	-3.8246	-99.99	0.0%	3
Heston–Merton	99.96	3.8235	-3.8235	-99.96	0.2%	2

RMSE% is 100×√mean(((model−market)/market)²). MAE and Bias $ are in option-premium dollars. Bias% is 100×mean((model−market)/market). Early-ex. frac. is the mean LSM share of paths exercised b

Page 10 · Table block · MSFT · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1-day lookback, hourly rolling, Δt = 1 min, n_paths=10000, seed=42. LSM horizon = remaining time to

Table MSFT-1 MSFT · Oct 2022 listed expiry 2022-10-21
2022-10-14 09:30 → 2022-10-14 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	44.10	4.4067	-4.4067	-41.28	85.2%	6
GARCH	43.95	4.4027	-4.4027	-41.24	91.2%	4
Heston	43.89	4.4108	-4.4108	-41.32	83.0%	2
Merton	44.01	4.3879	-4.3879	-41.10	88.2%	5
GARCH–Merton	**43.52**	**4.3457**	**-4.3457**	**-40.71**	**87.3%**	**1**
Heston–Merton	43.93	4.4167	-4.4167	-41.37	83.7%	3

Table MSFT-2 MSFT · Nov 2022 listed expiry 2022-11-18
2022-11-11 09:30 → 2022-11-11 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	96.77	5.9991	-5.9991	-96.76	8.5%	4
GARCH	97.47	6.0424	-6.0424	-97.46	7.0%	6
Heston	**93.95**	**5.8233**	**-5.8233**	**-93.92**	**11.8%**	**1**
Merton	96.66	5.9920	-5.9920	-96.64	9.2%	3
GARCH–Merton	97.09	6.0192	-6.0192	-97.08	8.1%	5
Heston–Merton	93.95	5.8234	-5.8234	-93.93	11.7%	2

Table MSFT-3 MSFT · Dec 2022 listed expiry 2022-12-16
2022-12-09 09:30 → 2022-12-09 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	22.48	1.3291	-1.3291	-19.99	94.1%	4
GARCH	23.49	1.3765	-1.3765	-20.70	96.7%	6
Heston	**17.27**	**1.0070**	**-1.0070**	**-15.14**	**76.4%**	**1**
Merton	22.40	1.3243	-1.3243	-19.92	94.1%	3
GARCH–Merton	22.99	1.3550	-1.3550	-20.38	95.8%	5
Heston–Merton	17.67	1.0120	-1.0120	-15.22	78.7%	2

Table MSFT-4 MSFT · Jan 2023 listed expiry 2023-01-20
2023-01-13 09:30 → 2023-01-13 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	24.91	1.7825	-1.7825	-22.63	95.6%	3
GARCH	25.10	1.7959	-1.7959	-22.80	96.4%	6
Heston	24.87	1.7725	-1.7725	-22.51	89.1%	2
Merton	25.03	1.7863	-1.7863	-22.68	95.4%	5
GARCH–Merton	24.98	1.7863	-1.7863	-22.68	98.2%	4
Heston–Merton	**24.85**	**1.7807**	**-1.7807**	**-22.61**	**92.8%**	**1**

RMSE% is 100×√mean(((model−market)/market)²). MAE and Bias $ are in option-premium dollars. Bias% is 100×mean((model−market)/market). Early-ex. frac. is the mean LSM share of paths exercised b

Page 11 · Table block · MSFT · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1-day lookback, hourly rolling, Δt = 1 min, n_paths=10000, seed=42. LSM horizon = remaining time to

Table MSFT-5 MSFT · Feb 2023 listed expiry 2023-02-17
2023-02-10 09:30 → 2023-02-10 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	99.29	7.3470	-7.3470	-99.28	2.3%	5
GARCH	99.10	7.3327	-7.3327	-99.09	2.3%	4
Heston	**96.73**	**7.1552**	**-7.1552**	**-96.69**	**5.7%**	**1**
Merton	99.31	7.3485	-7.3485	-99.30	2.2%	6
GARCH–Merton	99.09	7.3324	-7.3324	-99.09	3.0%	3
Heston–Merton	96.83	7.1629	-7.1629	-96.80	6.0%	2

Table MSFT-6 MSFT · Mar 2023 listed expiry 2023-03-17
2023-03-10 09:30 → 2023-03-10 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	95.62	4.9643	-4.9643	-95.47	7.4%	6
GARCH	94.94	4.9232	-4.9232	-94.68	6.9%	4
Heston	89.14	4.6065	-4.6065	-88.59	12.6%	2
Merton	95.60	4.9630	-4.9630	-95.44	8.2%	5
GARCH–Merton	94.45	4.8967	-4.8967	-94.17	7.3%	3
Heston–Merton	**88.96**	**4.5948**	**-4.5948**	**-88.36**	**12.7%**	**1**

Table MSFT-7 MSFT · Apr 2023 listed expiry 2023-04-21
2023-04-14 09:30 → 2023-04-14 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	98.40	9.7146	-9.7146	-98.38	7.4%	3
GARCH	99.24	9.7990	-9.7990	-99.23	4.8%	6
Heston	97.33	9.6064	-9.6064	-97.28	11.4%	2
Merton	98.42	9.7170	-9.7170	-98.40	8.7%	4
GARCH–Merton	98.77	9.7518	-9.7518	-98.75	6.7%	5
Heston–Merton	**97.24**	**9.5966**	**-9.5966**	**-97.18**	**10.1%**	**1**

Table MSFT-8 MSFT · May 2023 listed expiry 2023-05-19
2023-05-12 09:30 → 2023-05-12 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	**100.00**	**8.7750**	**-8.7750**	**-100.00**	**0.0%**	**1**
GARCH	100.00	8.7750	-8.7750	-100.00	0.0%	2
Heston	100.00	8.7750	-8.7750	-100.00	0.0%	3
Merton	100.00	8.7750	-8.7750	-100.00	0.0%	4
GARCH–Merton	100.00	8.7750	-8.7750	-100.00	0.0%	5
Heston–Merton	100.00	8.7750	-8.7750	-100.00	0.0%	6

RMSE% is $100\times\sqrt{\text{mean}(((\text{model}-\text{market})/\text{market})^2)}$. MAE and Bias $ are in option-premium dollars. Bias% is $100\times\text{mean}((\text{model}-\text{market})/\text{market})$. Early-ex. frac. is the mean LSM share of paths exercised b

Page 12 · Table block · MSFT · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1-day lookback, hourly rolling, Δt = 1 min, n_paths=10000, seed=42. LSM horizon = remaining time to

Table MSFT-9 MSFT · Jun 2023 listed expiry 2023-06-16
2023-06-09 09:30 → 2023-06-09 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	100.00	6.8000	-6.8000	-100.00	0.0%	3
GARCH	100.00	6.8000	-6.8000	-100.00	0.0%	4
Heston	**100.00**	**6.7998**	**-6.7998**	**-100.00**	**0.0%**	**1**
Merton	100.00	6.8000	-6.8000	-100.00	0.0%	5
GARCH–Merton	100.00	6.8000	-6.8000	-100.00	0.0%	6
Heston–Merton	100.00	6.7999	-6.7999	-100.00	0.0%	2

Table MSFT-10 MSFT · Jul 2023 listed expiry 2023-07-21
2023-07-14 09:30 → 2023-07-14 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	**100.00**	**10.4000**	**-10.4000**	**-100.00**	**0.0%**	**1**
GARCH	100.00	10.4000	-10.4000	-100.00	0.0%	2
Heston	100.00	10.4000	-10.4000	-100.00	0.0%	3
Merton	100.00	10.4000	-10.4000	-100.00	0.0%	4
GARCH–Merton	100.00	10.4000	-10.4000	-100.00	0.0%	5
Heston–Merton	100.00	10.4000	-10.4000	-100.00	0.0%	6

Table MSFT-11 MSFT · Aug 2023 listed expiry 2023-08-18
2023-08-11 09:30 → 2023-08-11 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	**100.00**	**6.9500**	**-6.9500**	**-100.00**	**0.0%**	**1**
GARCH	100.00	6.9500	-6.9500	-100.00	0.0%	2
Heston	100.00	6.9500	-6.9500	-100.00	0.0%	3
Merton	100.00	6.9500	-6.9500	-100.00	0.0%	4
GARCH–Merton	100.00	6.9500	-6.9500	-100.00	0.0%	5
Heston–Merton	100.00	6.9500	-6.9500	-100.00	0.0%	6

Table MSFT-12 MSFT · Sep 2023 listed expiry 2023-09-15
2023-09-08 09:30 → 2023-09-08 15:59 · n = 6 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	**100.00**	**9.6000**	**-9.6000**	**-100.00**	**0.0%**	**1**
GARCH	100.00	9.6000	-9.6000	-100.00	0.0%	2
Heston	100.00	9.6000	-9.6000	-100.00	0.0%	3
Merton	100.00	9.6000	-9.6000	-100.00	0.0%	4
GARCH–Merton	100.00	9.6000	-9.6000	-100.00	0.0%	5
Heston–Merton	100.00	9.6000	-9.6000	-100.00	0.0%	6

RMSE% is 100×√mean(((model−market)/market)²). MAE and Bias $ are in option-premium dollars. Bias% is 100×mean((model−market)/market). Early-ex. frac. is the mean LSM share of paths exercised b

Summary tables · RMSE% across companies and twelve windows

Table S1 · Overall ranking (mean RMSE% over 36 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 36 cells bes
1	**Heston**	**89.16**	**99.97**	**7**
2	Heston–Merton	89.17	99.97	9
3	GARCH–Merton	90.17	100.00	6
4	GBM	90.27	100.00	12
5	Merton	90.27	100.00	1
6	GARCH	90.37	100.00	1

Table S2 · Wins by company (lowest RMSE% in that company×window

Model	SPY wins	AAPL wins	MSFT wins	Total
GBM	**6**	**2**	**4**	**12**
GARCH	0	1	0	1
Heston	0	3	4	7
Merton	0	1	0	1
GARCH–Merton	3	2	1	6
Heston–Merton	3	3	3	9

Table S3 · Mean RMSE% by window (equal-weight mean of SPY / AAPL / MSFT)

Model	Oct22	Nov22	Dec22	Jan23	Feb23	Mar23	Apr23	May23	Jun23	Jul23	Aug23	Sep23	Mean of 12
GBM	52.89	98.92	68.31	74.91	99.22	90.69	98.30	100.00	100.00	100.00	100.00	100.00	90.27
GARCH	53.25	99.11	68.75	74.92	99.27	90.17	98.92	100.00	100.00	100.00	100.00	100.00	90.37
Heston	**52.25**	**97.97**	**64.28**	**74.86**	**97.22**	**85.44**	**97.97**	**100.00**	**99.98**	**100.00**	**100.00**	**99.99**	**89.16**
Merton	52.88	98.88	68.31	74.96	99.20	90.75	98.31	100.00	100.00	100.00	100.00	100.00	90.27
GARCH–Mertor	52.42	98.99	68.22	74.86	99.21	89.65	98.64	100.00	100.00	100.00	100.00	100.00	90.17
Heston–Mertor	52.49	97.97	64.40	74.87	97.20	85.26	97.91	100.00	99.98	100.00	100.00	99.99	89.17

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×window wins. A model can win the most cells without having the lowest average error (and vice versa).

Summary by company · mean RMSE% across twelve windows

Table S4.1 · SPY

Model	Mean RMSE%	Median	# windows bes
GBM	94.45	100.00	6
GARCH	94.49	100.00	0
Heston	94.24	100.00	0
Merton	94.47	100.00	0
GARCH–Merton	94.38	100.00	3
Heston–Merto	**94.21**	**100.00**	**3**

Table S4.2 · AAPL

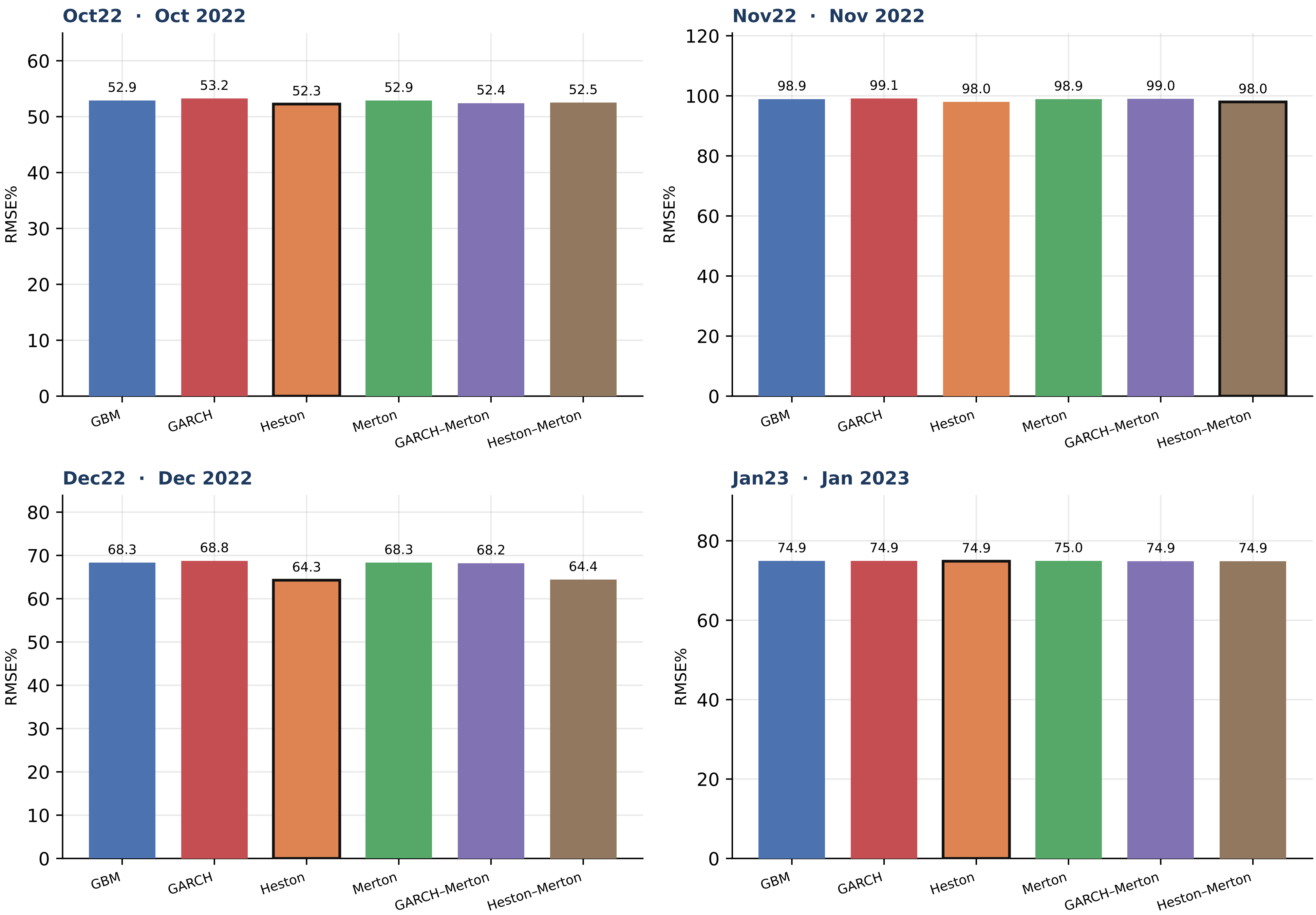
Model	Mean RMSE%	Median	# windows bes
GBM	94.56	99.91	2
GARCH	94.67	99.77	1
Heston	**92.98**	**99.82**	**3**
Merton	94.57	99.91	1
GARCH–Merton	94.38	99.73	2
Heston–Merton	93.01	99.84	3

Table S4.3 · MSFT

Model	Mean RMSE%	Median	# windows bes
GBM	81.80	98.84	4
GARCH	81.94	99.17	0
Heston	**80.26**	**97.03**	**4**
Merton	81.79	98.86	0
GARCH–Merton	81.74	98.93	1
Heston–Merton	80.29	97.04	3

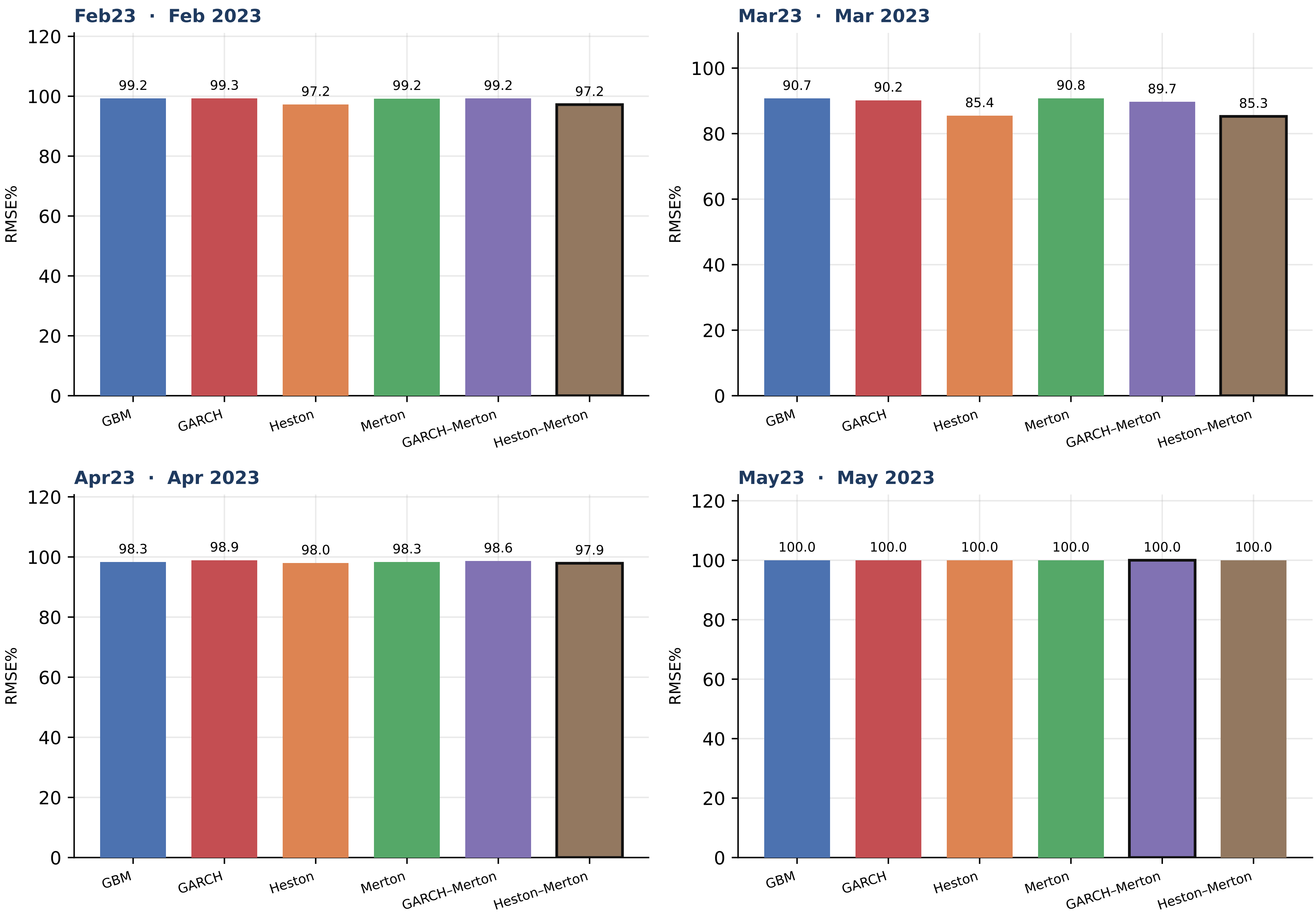
Each column is one company. Green row = lowest mean RMSE% across the twelve Friday-before-expiry windows for that name.

Figure 1 · RMSE% by model and window · mean of SPY / AAPL / MSFT



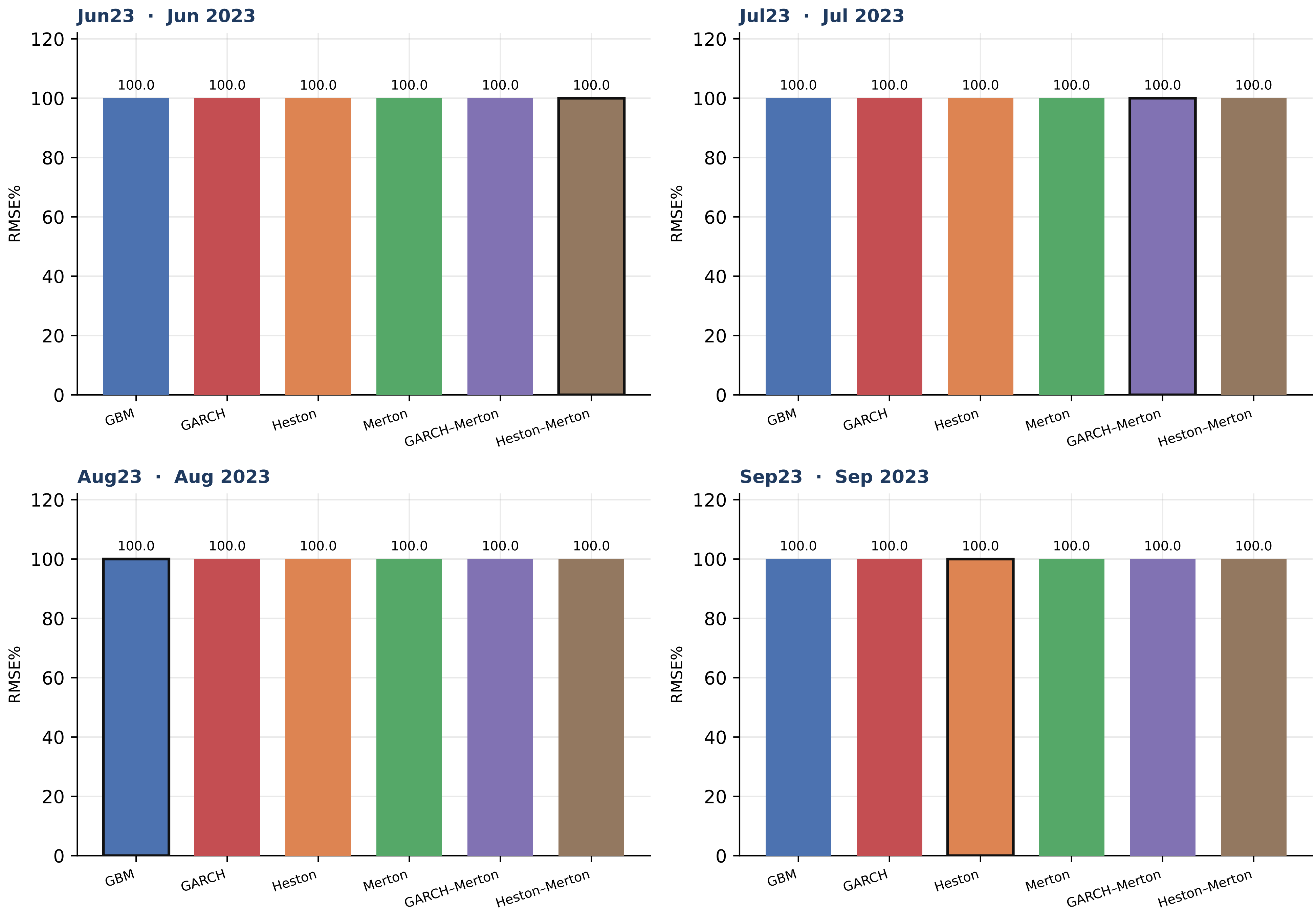
Outlined bar = lowest mean RMSE% in that window. Equal-weight mean of the three companies.

Figure 1 · RMSE% by model and window · mean of SPY / AAPL / MSFT



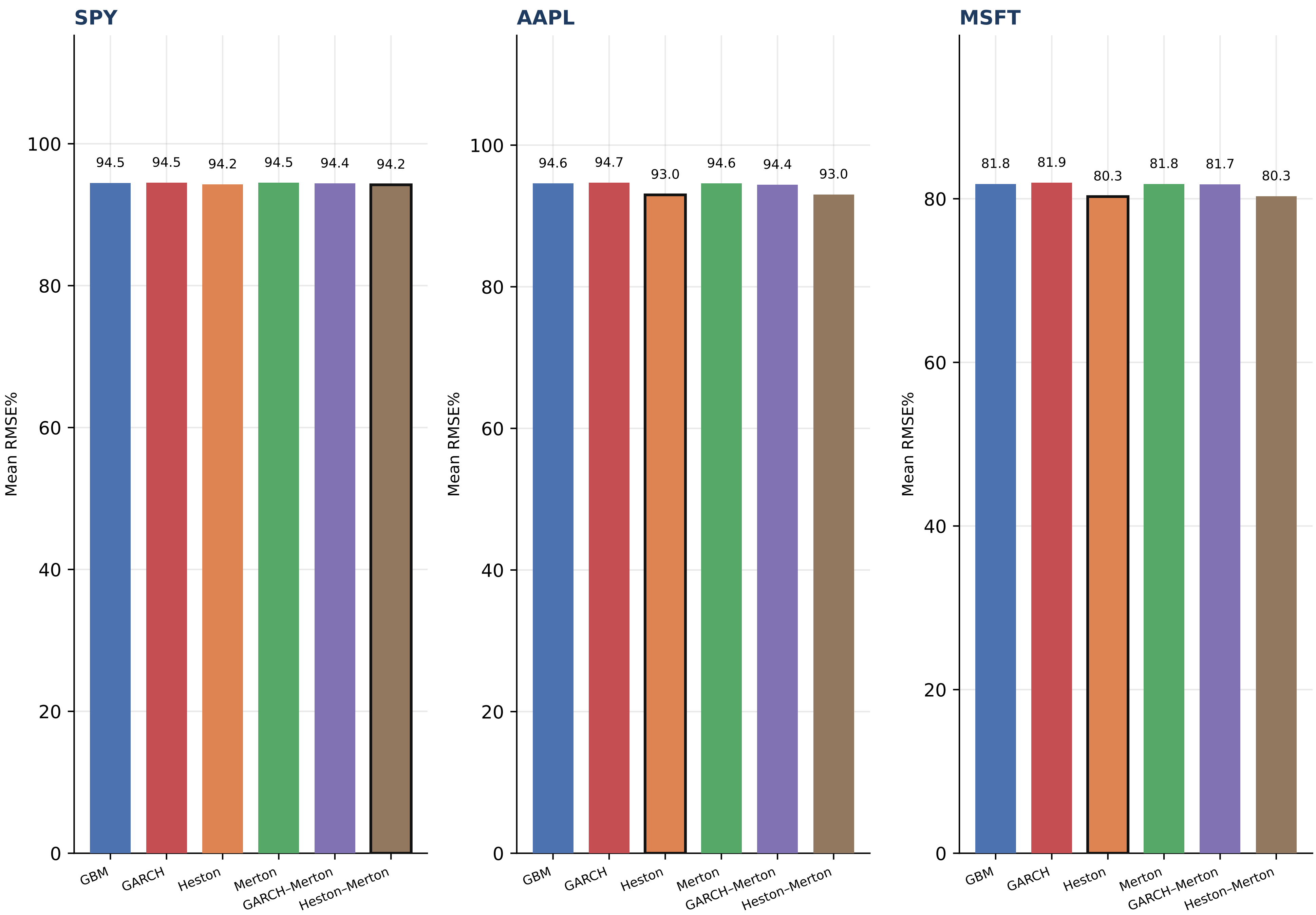
Outlined bar = lowest mean RMSE% in that window. Equal-weight mean of the three companies.

Figure 1 · RMSE% by model and window · mean of SPY / AAPL / MSFT



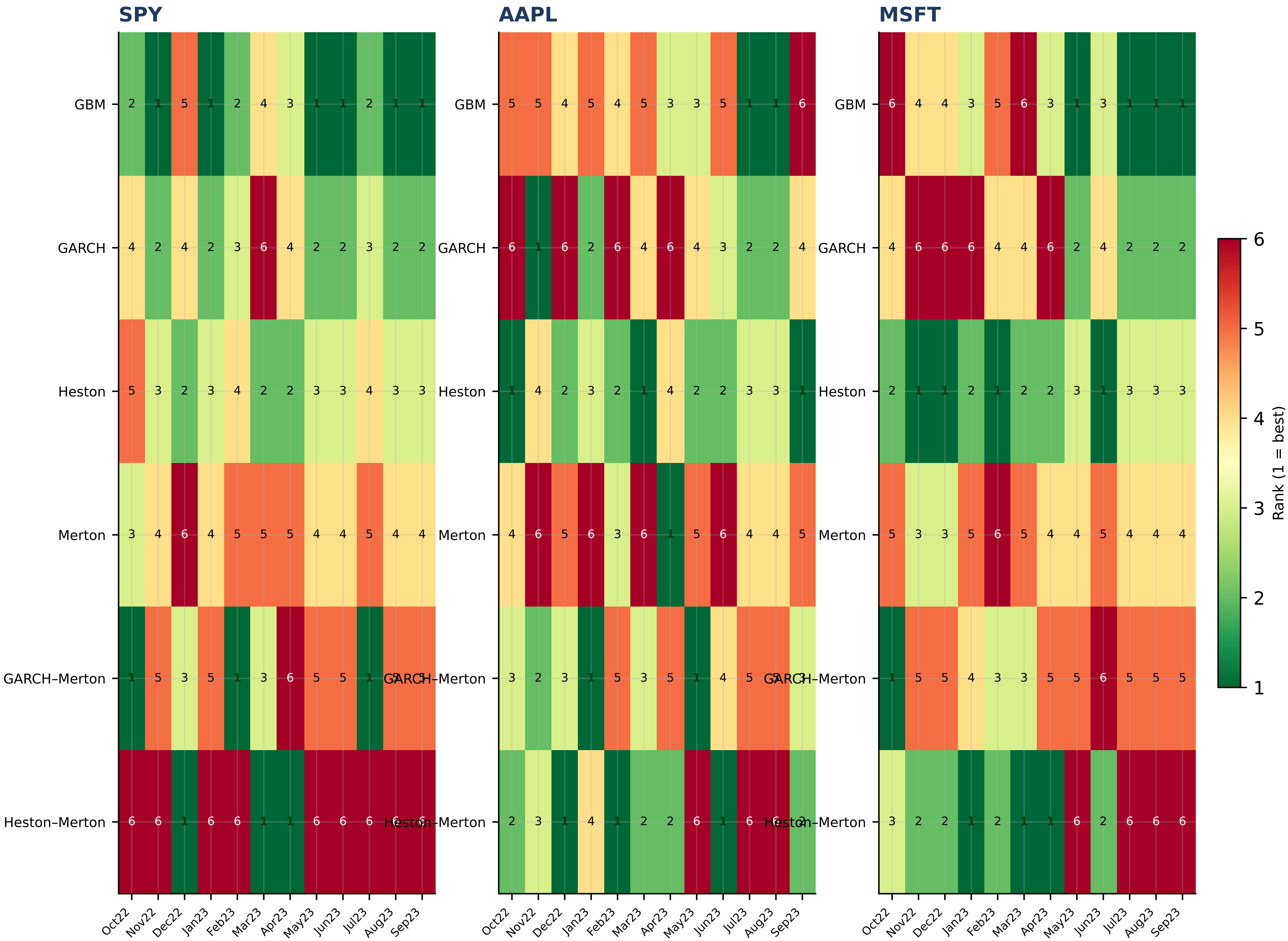
Outlined bar = lowest mean RMSE% in that window. Equal-weight mean of the three companies.

Figure 2 · RMSE% by model · each company (mean across twelve windows)



Outlined bar = lowest mean RMSE% for that company.

Figure 3 · Model rank by company × window (1 = best RMSE%)



Green / 1 = lowest RMSE% in that company×window cell. Rankings can and do change across names and windows.

Final analysis

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 36 company×window cells, Heston is the best overall model (mean RMSE% = 89.16; 7 of 36 cells).
- Second overall is Heston–Merton (mean RMSE% = 89.17; 9 cells).
- The ranking is not stable. Best-in-cell models across the 36 tables: GARCH, GARCH–Merton, GBM, Heston, Heston–Merton, Merton.
- SPY: lowest mean RMSE% is Heston–Merton (94.21); wins 3 of 12 windows.
- AAPL: lowest mean RMSE% is Heston (92.98); wins 3 of 12 windows.
- MSFT: lowest mean RMSE% is Heston (80.26); wins 4 of 12 windows.
- Jump models as a group have lower mean RMSE% (89.87) than the three no-jump models (89.93).
- These conclusions are conditional on the V3 filters, 1-day hourly calibration, nearest-ATM hourly sample, LSM with 10000 paths and 1-minute steps to Friday 15:59, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 36 detailed tables

- Each of Tables SPY-1…MSFT-12 uses exactly the same option contracts for all six models. Differences are the dynamics, not the sample.
- Mark 1 is always lowest RMSE%. MAE, Bias $, Bias%, and early-exercise are reported, not used for the ranking.
- A model that is Mark 1 on RMSE% can still be biased (systematically expensive or cheap).
- Early-exercise fractions are a model implication, not an accuracy score; LSM stops at window Friday close, not listed expiry.